Question 1:

In a car race, car *A* takes 4 s less than car *B* at the finish and passes finishing line with a speed *V* more than car *B*. Assuming that the cars start from rest and travel with constant accelerations of 4 m/s^2 and 1 m/s^2 respectively, then the value of *V* in (m/s) is

- A. 12
- B. 8
- C. $\frac{4}{3}$
- D. 5

Question 2:

The position of a particle moving along y-axis is given as $y = 3t - t^2$, where y is in metre and t is in second. Magnitude of acceleration of the particle at $t = 8$ s is

- A. 2 m/s^2
- B. 6 m/s^2
- C. 1 m/s^2
- D. 3 m/s^2

Question 3:

A 150 m long train is moving with uniform velocity of 54 km/h. The time taken by train to cross a bridge of length 600 m will be

- A. 56 s
- B. 68 s
- C. 50 s
- D. 40 s

Question 4:

When a small sphere moves through a fluid, the viscous force F opposing the motion is found experimentally to depend on radius of sphere r, velocity of sphere v and viscosity η of the fluid such that, $F = K\eta^a r^b v^c$, where K is dimensionless constant. The value of term $b - c$ is equal to

- A. Zero
- B. 1
- C. 2
- D. 3

Question 5:

In a new system of units, unit of mass is 10 kg, unit of length is 10 m and unit of time is 10 s. What will be the potential energy of body in this new system if in S.I. system its potential energy is 30 J?

- A. 2 units
- B. 3 units
- C. 20 units
- D. 300 units

If a physical quantity depends on mass m and time T as $K = \frac{4\pi^2 m}{T^2}$ and percentage error in measurement of mass (m) is 1% while percentage error in measurement of time (T) is 2%, then the maximum percentage error in measurement of K is

A. 4%

B. 3%

C. 5%

D. 2%

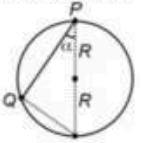
Question 7:

A motor boat can row in still water at a speed of 10 m/s. There is a small island in a river flowing at 5 m/s. The boat starts from island and goes downstream to a temple that is 100 m away. The boat then returns back to the island. The time taken by the boat to complete round trip is

- A. 26.7 s
- B. 24.3 s
- C. 36.2 s
- D. 27.6 s

Question 8:

A frictionless wire PQ is fixed on a circular ring of radius R as shown in the figure. A small spherical bead slips on this wire when it is released from P, the time taken by ball to slip from P to Q is



A. $2\sqrt{\frac{R}{g\cos\alpha}}$

B. $2\sqrt{\frac{R\cos\alpha}{g}}$

C. $2\sqrt{\frac{R}{g}}$

D. $2\sqrt{\frac{R}{g\sin\alpha}}$

Question 9:

The number of significant zeroes in measured value 0.0052300 is

- A. Four
- B. Two
- C. Five
- D. Three

Question 10:

If, $X = 0.72 \times 3.87$, then the number of significant digits in value of X will be

- A. Three
- B. Two
- C. One
- D. Four

Question 11:

Which among the following is a supplementary unit?

- A. kg
- B. radian
- C. s
- D. m

Question 12:

Consider the following statement out of which one is labelled as **Assertion (A)** and other as **Reason (R)**.
Assertion (A): If two measuring devices have different precision, the one with higher precision will always yield more accurate results.
Reason (R): A more precise instrument has always higher errors.
In the light of above statements choose the correct option.

- A. Both (A) and (R) are true but (R) is not the correct explanation of (A)
- B. Both (A) and (R) are true and (R) is the correct explanation of (A)
- C. (A) is true but (R) is false
- D. Both (A) and (R) are false

Question 13:

Two paper screens *A* and *B* are separated by certain distance *D*. The bullet first pierces *A* and then *B*. The hole in *B* is 15 cm below the hole in *A*. If the bullet was travelling horizontally with a speed of $500\sqrt{3}\ m/s$ at time of hitting *A*, then the value of *D* is ($g = 10\ m/s^2$)

- A. 150 m
- B. 3000 m
- C. 500 m
- D. $500\sqrt{3}\ m$

Question 14:

A projectile is launched at an angle α (upward) from horizontal with velocity 40 m/s from a large height. After 10 s, its inclination with horizontal is β (upward), then value of $\tan\beta$ is ($g = 10\ m/s^2$)

- A. $\tan\alpha + 5\sec\alpha$
- B. $\tan\alpha - 5\sec\alpha$
- C. $\tan\alpha - 2.5\sec\alpha$
- D. $\tan\alpha + 2.5\sec\alpha$

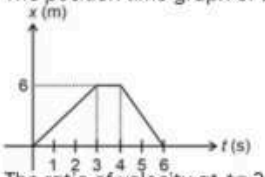
Question 15:

A ball projected (at $t = 0$) vertically up passes the same height h, at $t = 2$ s and at $t = 10$ s. The value of h is equal to (Take $g = 10\ m/s^2$)

- A. 102 m
- B. 100 m
- C. 98 m
- D. 90 m

Question 16:

The position time graph of an object moving along x-axis is shown.



The ratio of velocity at $t = 2$ s to the velocity at $t = 5$ s is

A. $-\frac{3}{2}$

B. $\frac{2}{1}$

C. $\frac{1}{2}$

D. $\frac{-2}{3}$

Question 17:

Consider a car moving with a velocity of 54 km/h. At a distance of 450 m from traffic light brakes are applied. The acceleration of car after brakes are applied is $-0.3\ m/s^2$. The car's position relative to traffic light when it stops will be

- A. 375 m
- B. 425 m
- C. 75 m
- D. 25 m

Question 18:

A physical quantity x depends on quantities y and z as follows, $x = Ay + B\cos(Cz)$, where, A, B and C are dimensional constants. Which of the following have same dimension?

- A. C and z^{-1}
- B. C and z
- C. x and B
- D. Both (1) and (3)

Question 19:

The dimensions of voltage are expressed as $[V] = [M^X L^Y T^Z A^P]$, then the value of x is

A. 2

B. −3

C. −1

D. 1

Question 20:

If, $L = 3.331$ cm and $B = 3.1$ cm, then value of $L + B$, taking significant figure into account, will be

A. 6.431 cm

B. 6.43 cm

C. 6.4 cm

D. 6 cm

Question 21:

The speed with which a ball should be projected vertically upwards, so that distance covered by it in 4th second is twice the distance it covers in 5th second, is

A. 55 m/s

B. 65 m/s

C. 70 m/s

D. 85 m/s

Question 22:

A ball is dropped from a certain height such that distance travelled in first five seconds is equal to distance covered in last second of its motion. The time taken by ball to reach ground is

A. 12 s

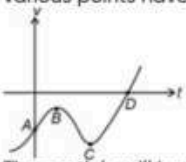
B. 10 s

C. 15 s

D. 13 s

Question 23:

A particle moves in a straight line. The velocity (v) varies with time (t) as shown and various points have been marked on graph.



The particle will have zero acceleration at point.

A. A

B. B

C. C

D. Both (2) and (3)

Question 24:

The motion of a body is governed by the equation, $a = 5 - 4v$, where a denotes the acceleration (in m/s^2) and v denotes the velocity (in m/s). If body was initially at rest, then magnitude of initial acceleration is

A. 5 m/s^2

B. Zero

C. 3 m/s^2

D. 4 m/s^2

Question 25:

A particle moves along a circle of radius 1 m. The displacement of particle when it has traversed half of circle is

A. Zero

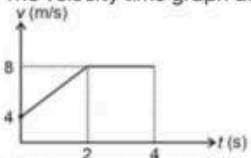
B. 1 m

C. 2 m

D. 0.5 m

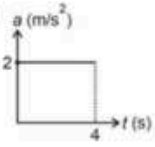
Question 26:

The velocity time graph describing motion of a bicycle is shown.

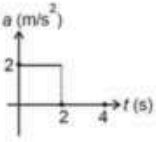


The corresponding acceleration time graph will be

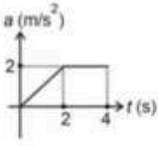
A.



B.



C.



D.



Question 27:

A particle is projected from horizontal plane ($x - z$) (where y-axis is along vertical) such that its velocity at any time t is given as, $\vec{v} = \alpha\hat{i} + \left(\beta - \gamma t\right)\hat{j}$. The maximum height attained by the particle will be

A. $\frac{\alpha^2}{2\gamma}$

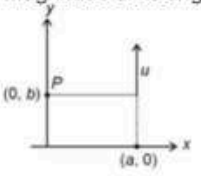
B. $\frac{\beta^2}{2\gamma}$

C. $\frac{2\alpha^2}{\gamma}$

D. $\frac{2\beta^2}{\gamma}$

Question 28:

A particle is moving with constant speed u in x-y plane as shown in figure. The magnitude of its angular velocity about point P at the given instant is



A. Zero

B. $\frac{u}{a}$

C. $\frac{u}{b}$

D. $\frac{u}{\sqrt{a^2+b^2}}$

Question 29:

The magnitude of resultant of two unit vectors is also a unit vector. The angle between the two vectors will be equal to

A. $\frac{\pi}{3}$

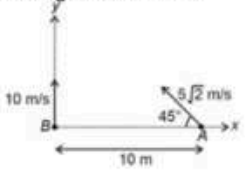
B. $\frac{\pi}{2}$

C. $\frac{2\pi}{3}$

D. $\frac{\pi}{4}$

Question 30:

Two projectiles *A* and *B* are projected simultaneously in a horizontal *x*-*y* plane from two different points on *x*-axis as shown. The minimum distance between these two particles during their motion is



A. 10 m

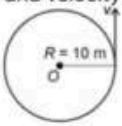
B. $5\sqrt{2}\,m$

C. $\frac{5}{\sqrt{2}}\,m$

D. $10\sqrt{2}\,m$

Question 31:

A car moves on a circular path of radius 10 m. At a certain instant as shown speed of car is 10 m/s and it is decreasing at a rate of 5 m/s^2. The angle between acceleration vector and velocity vector at this instant is



A. $\tan^{-1}\left(\frac{1}{2}\right)$

B. $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{2}\right)$

C. $\frac{\pi}{2}-\tan^{-1}(2)$

D. $\frac{\pi}{2}+\tan^{-1}(2)$

Question 32:

The x and y coordinates of the particle at any time are, $x = 5t - 2t^2$ and $y = 10t$ respectively, where x and y are in meters and t in seconds.
Based on above information, consider following statements and choose the correct option.
A. The motion of the particle is uniformly accelerated.
B. The motion of the particles is non-uniformly accelerated.
C. The trajectory of the particle will be circular.
D. The trajectory of the particle will be straight line.

- A. Both statements A and B are correct
- B. Both statements C and D are correct
- C. Only statement A is correct
- D. All statements are correct

Question 33:

Consider the following statements.
A. It is possible for a body to have a constant velocity while the speed of body is varying.
B. A body having an acceleration of constant magnitude can move with constant speed.
C. A body starting from rest cannot move in straight line if it experiences acceleration varying in magnitude.
Based on above information, choose the correct option.

- A. Only statement A is correct
- B. Both statement B and C are correct
- C. Only statement B is correct
- D. All statements are correct

Question 34:

Consider a vector $\vec{A}$ given by, $\vec{A} = 4\hat{i} + 3\hat{j} + 5\hat{k}$. The angle made by the vector with y-axis is

A. $\cos^{-1}\left(\frac{3}{5\sqrt{2}}\right)$

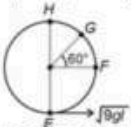
B. $\cos^{-1}\left(\frac{4}{5\sqrt{2}}\right)$

C. $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$

D. $\cos^{-1}\left(\frac{5}{3\sqrt{2}}\right)$

Question 35:

A particle is suspended in vertical plane with a string of length l and it is given horizontal speed, $u = \sqrt{9gl}$ at the bottom as shown.



In column I, magnitude of tangential acceleration at various points are asked and in column II the value of acceleration is given. Match entries in column I with entries in column II

	Column-I		Column-II
a.	Tangential acceleration at point E	(p)	$5\ m/s^2$
b.	Tangential acceleration at point F	(q)	Zero
c.	Tangential acceleration at point G	(r)	$10\ m/s^2$
d.	Tangential acceleration at point H	(s)	$5\sqrt{3}\ m/s^2$

A. a(q), b(r), c(p), d(q)

B. a(q), b(q), c(s), d(q)

C. a(q), b(r), c(s), d(q)

D. a(q), b(r), c(r), d(p)

Question 36:

Identify the vector quantity among the following

A. Speed

B. Energy

C. Velocity

D. Heat

Question 37:

For angles of projection of a projectile at angles α and 90° − α, the horizontal ranges described by projectile are in ratio of (Assume the speed of projection to be same and projected from the ground).

A. 1 : 1

B. 2 : 3

C. 1 : 2

D. 2 : 1

Question 38:

Two non-zero vectors $\vec{P}$ and $\vec{Q}$ are such that, $\left|\vec{P}+2\vec{Q}\right|=\left|P-2\vec{Q}\right|$. The angle between the two vectors will be

A. $\frac{\pi}{2}$

B. $\frac{\pi}{3}$

C. $\frac{3\pi}{4}$

D. $\frac{\pi}{4}$

Question 39:

Given below are two statements:
Statement I: A vector quantity has both, magnitude and direction and it must follow triangle law of addition.

Statement II: If we multiply a vector $\vec{A}$ with a number λ, then $\left|\lambda\vec{A}\right| = \lambda\left|\vec{A}\right|$.

In the light of the above statements, choose the most appropriate answer from the options given below.

A. Statement I is correct but statement II is incorrect

B. Both statement I and statement II are correct

C. Both statement I and statement II are incorrect

D. Statement I is incorrect but statement II is correct

Question 40:

A vehicle travels half the distance with speed $2v$ and the remaining distance with speed $4v$. Its average speed is

A. $\frac{v}{2}$

B. $\frac{3v}{4}$

C. $\frac{8v}{3}$

D. v

Question 41:

The numerical value of the ratio $\frac{|\text{Displacement}|}{\text{Distance}}$ is

A. Always less than or equal to one

B. Always greater than or equal to one

C. Always equal to one

D. May be greater than one

Question 42:

The length (ℓ) measured by a measuring instrument is represented as, $\ell = (4.9 \pm 0.1)$ m, then

A. $\ell > 5$ m

B. $\ell < 4.8$ m

C. $4.8\text{ m} \leq \ell \leq 5\text{ m}$

D. $\ell < 0.1$ m

Question 43:

The dimensional formula for most probable velocity of gas molecules is given by

A. $[M^0L^0T^0]$

B. $[M^0L^0T^{-1}]$

C. $[M^1L^0T^{-1}]$

D. $[M^0L^1T^{-1}]$

Question 44:

The average value of cos$x$ in the interval $x = 0$ to $x = \pi$ will be

A. Zero

B. $\frac{1}{\sqrt{2}}$

C. $\frac{2}{\pi}$

D. $\frac{1}{2}$

Question 45:

The refractive index of a medium depends upon the wavelength (λ) of light as,
$\mu = A + \frac{B}{\lambda^2}$
The dimensions of $\frac{A}{B}$ is

A. $[M^0L^0T^0]$

B. $[M^0L^{-2}T^0]$

C. $[M^0L^2T^0]$

D. $[M^0L^{-2}T^{-1}]$

Question 46:

Consider the following reaction:
$A + B + 2C \rightarrow ABC_2$
For the above reaction 5 g of B, 6.02×10^{22} atoms of A and 0.02 mol of C are taken and the amount of ABC_2 formed is 2.2 g. If atomic masses of A and B are 80 u and 40 u respectively then the atomic mass of C will be

A. 90 u

B. 60 u

C. 100 u

D. 50 u

Question 47:

An element E has 2 isotopes with their natural abundances in the ratio of 1 : 4. If the atomic mass of isotope with lesser % abundance is 18 u and the average atomic mass of the element is 19.6 u then the atomic mass of other isotope will be

A. 21 u

B. 20 u

C. 19.2 u

D. 18.6 u

Question 48:

If 0.05 mol of Barium chloride reacts with 14.2 g of sodium sulphate then the amount of excess reagent left after reaction is

A. 4.2 g

B. 6.3 g

C. 7.1 g

D. 8.2 g

Question 49:

The number of atoms present in the gaseous product obtained after complete combustion of 11.6 g of n-butane (C_4H_{10}) at room temperature is nearly

A. 4.82×10^{24}

B. 1.44×10^{23}

C. 4.82×10^{23}

D. 1.44×10^{24}

Question 50:

Match the column I with column II.

	Column-I (Prefix)		Column-II (Multiple)
a.	Zeta	(i)	10^{9}
b.	Peta	(ii)	10^{18}
c.	Exa	(iii)	10^{15}
d.	Giga	(iv)	10^{21}

Choose the correct match.

A. a(iii), b(iv), c(ii), d(i)

B. a(iii), b(iv), c(i), d(ii)

C. a(iv), b(iii), c(ii), d(i)

D. a(iv), b(iii), c(i), d(ii)

Question 51:

The ratio of total number of electrons in 4.8 g of methane gas to 6.4 g of oxygen gas at STP is

A. 3 : 4

B. 15 : 16

C. 16 : 19

D. 2 : 3

Question 52:

A sample of aluminium sulphate contains 0.3 mole of sulphur atoms. The number of mole(s) of oxygen atoms in the sample is

A. 0.3

B. 0.6

C. 0.9

D. 1.2

Question 53:

Consider the following statements:
a. Precision refers to the agreement of a particular value to the true value.
b. 0.0034 has four significant figures.
c. 1.002 has four significant figures
d. 1×10^2 has three significant figures.
e. The sum of 13.12, 16.0 and 1.024 in correct significant figures is 30.1.
Choose the option with all incorrect statements.

A. a, b and d only

B. a, d and e only

C. b, d and e only

D. c and e only

Question 54:

How many grams of concentrated nitric acid solution should be used to prepare 450 mL of 4.0 M HNO_3 solution?
[Concentrated HNO_3 is 70% (w/w) HNO_3]

A. 148 g

B. 162 g

C. 221 g

D. 249 g

Question 55:

In which of the given compounds, the mass percentage of oxygen is highest?

A. H_2SO_4

B. Na_2SO_4

C. $CaCO_3$

D. Al_2O_3

Question 56:

When 100 mL of CO_2 gas at STP is passed over red hot coke, the volume of gaseous mixture after complete reaction at STP becomes 180 mL. Consider the following statements:
a. Volume of CO obtained is 160 mL
b. Total number of oxygen atoms present in gaseous mixture after reaction is 0.0115 N_A
c. Moles of CO obtained at STP are $\frac{1}{140}$ mol
d. Volume of CO_2 left after reaction is 80 mL
Choose the correct statement(s).

A. a and b only

B. a and c only

C. d only

D. a, b, c and d

Question 57:

If an enzyme contains 0.02% by weight of magnesium then its minimum molar mass will be

A. 200000 g mol^{-1}

B. 6000 g mol^{-1}

C. 120000 g mol^{-1}

D. 240000 g mol^{-1}

Question 58:

Quantum numbers of four different electrons of an atom are given below.

a. $n = 4,\ l = 2,\ m_l = -2,\ m_s = -\frac{1}{2}$

b. $n = 3,\ l = 2,\ m_l = -1,\ m_s = +\frac{1}{2}$

c. $n = 3,\ l = 1,\ m_l = +1,\ m_s = -\frac{1}{2}$

d. $n = 4,\ l = 1,\ m_l = 0,\ m_s = +\frac{1}{2}$

The correct order of increasing energies is

A. d < a < c < b

B. d < c < b < a

C. b < c < d < a

D. c < b < d < a

Question 59:

An ion with mass number 37 possesses one unit negative charge. If the ion contains 11.1% more neutrons than the electrons then the atomic number of the ion is

A. 16

B. 17

C. 18

D. 19

Question 60:

A laser light produces a photon of a wavelength 355 nm. If the number of photons emitted per second is 3.55×10^{24}, then the power of this laser light will be ($h = 6 \times 10^{-34}$ Js)

A. 9×10^{6} watt

B. 1.8×10^{5} watt

C. 9×10^{5} watt

D. 1.8×10^{6} watt

Question 61:

Consider the following statements:
a. The number of angular nodes in 3*p* orbital is 1.
b. The number of radial nodes in 4*s* orbital is 2.
c. The total number of nodes in 5*p* orbital is 3.
d. The number of radial nodes in 2*s* and 3*p* are same.
Choose the option containing correct statements

A. a and d only

B. b and c only

C. b, c and d only

D. c and d only

Question 62:

Choose the pair of *d*-orbitals in which electron density is aligned along the axes?

A. d_{xy} and d_{z^2}

B. d_{xy} and d_{zx}

C. d_{yz} and $d_{x^2-y^2}$

D. $d_{x^2-y^2}$ and d_{z^2}

Question 63:

Match the following columns

	Column I (Set of quantum numbers (n and l) representing subshell		Column II (Maximum number of electrons accommodated)
a.	n = 3, l = 2	(i)	6
b.	n = 2, l = 1	(ii)	14
c.	n = 4, l = 3	(iii)	2
d.	n = 1, l = 0	(iv)	10

Choose the correct match.

A. a(ii), b(i), c(iv), d(iii)

B. a(iv), b(i), c(ii), d(iii)

C. a(iv), b(iii), c(i), d(ii)

D. a(ii), b(iii), c(iv), d(i)

Question 64:

A particular station of All India Radio, New Delhi broadcasts on a frequency of 3150 kHz. Choose the incorrect statement(s) from the following.
a. The wavelength is 95.24 m.
b. The wavenumber associated with the given frequency is $1.5 \times 10^{-3}\ m^{-1}$.
c. Total energy associated with the given frequency will be 2.087×10^{-27} J.
Choose the correct option.

A. a and c only

B. b only

C. b and c only

D. a, b and c

Question 65:

Choose the correct statement from the following.

A. Diffraction and interference are the phenomena which can be explained by particle nature of the electromagnetic radiation

B. Electromagnetic waves require a medium to propagate

C. The frequency of radio waves is higher than that of microwaves

D. The oscillating electric and magnetic fields produced by oscillating charged particles are perpendicular to each other

Question 66:

Consider an electron present in ground state of a hydrogen like species with Z = 4 and match the column I with column II

	Column I		Column II
a.	Velocity of electron in ground state (m/s)	(i)	3.48×10^{-17}
b.	Orbit angular momentum	(ii)	1.32×10^{-11}
c.	Radius of ground state in metre	(iii)	1.05×10^{-34}
d.	Magnitude of energy of ground state in joules	(iv)	8.72×10^{6}

A. a(iii), b(i), c(iv), d(ii)

B. a(iii), b(iv), c(ii), d(i)

C. a(iv), b(iii), c(ii), d(i)

D. a(iv), b(iii), c(i), d(ii)

Question 67:

A ball of mass 6.62×10^5 mg is moving with a speed of 100 m/s. The associated wavelength will be
($h = 6.62 \times 10^{-34}$ Js)

A. 1×10^{-35} m

B. 1×10^{-39} m

C. 1×10^{-37} m

D. 1×10^{-30} m

Question 68:

If uncertainty in position is 3 times of the uncertainty in momentum then the uncertainty in velocity is

A. $\frac{1}{2m}\sqrt{\frac{h}{3\pi}}$

B. $\frac{1}{3m}\sqrt{\frac{h}{2\pi}}$

C. $\frac{h}{\sqrt{6\pi m}}$

D. $\sqrt{\frac{h}{6\pi m^2}}$

Question 69:

Which of the given species is isoelectronic with CN^-?

A. O_2^+

B. N_2^-

C. CO

D. NO

Question 70:

Match the given column I containing species with column II containing the number of unpaired electrons in d-subshell.

	Column-I		Column-II
a.	Chromium	(i)	2
b.	Iron	(ii)	Zero
c.	Nickel	(iii)	5
d.	Copper	(iv)	4

Choose the correct match

A. a(iii), b(i), c(iv), d(ii)

B. a(iii), b(iv), c(i), d(ii)

C. a(iv), b(i), c(ii), d(iii)

D. a(iv), b(iii), c(i), d(ii)

Question 71:

Consider the following statements regarding photoelectric effect
a. The number of photoelectrons ejected is proportional to the frequency of light used.
b. There is no time lag between the striking of beam of light and the ejection of electrons from the metal surface.
c. For each metal, there is a characteristic maximum frequency above which photoelectric effect is not observed.
d. The work function of sodium metal is less than that of potassium.
e. The kinetic energy of the photoelectrons increases with the increase in intensity of light.
Choose the option containing only incorrect statements.

A. a, b, c, d and e

B. b, c and d only

C. a, b and d only

D. a, c, d and e only

Question 72:

Consider the following statements regarding the Bohr's atomic model.
a. It fails to account for the finer details of the spectrum of hydrogen atom observed by using sophisticated spectroscopic techniques.
b. Bohr's theory could not explain the ability of atoms to form molecules by chemical bonds.
c. Bohr's theory was unable to explain Zeeman effect.
Choose the correct statements.

A. a and b only

B. b and c only

C. a and c only

D. a, b and c

Question 73:

All of the following transitions in the spectrum of sample of hydrogen atom fall in IR region, except

A. $n = 4$ to $n = 2$

B. $n = 5$ to $n = 3$

C. $n = 4$ to $n = 3$

D. $n = 5$ to $n = 4$

Question 74:

Select the correct statements from the following.
a. In the absence of electrical or magnetic field, cathode rays travel in straight line.
b. Mass of electron is nearly 9.1×10^{-31} kg
c. Protons and electrons are collectively known as nucleons.
d. All the isotopes of a given element show similar chemical properties.
Choose the correct answer from the options given below.

A. a and b only

B. a and c only

C. a, b and d only

D. a, c and d only

Question 75:

The de-Broglie wavelength of a wave associated with a moving subatomic particle with kinetic energy K is λ. The wavelength of particle, if its kinetic energy becomes 4K is

A. 2λ

B. $\frac{\lambda}{2}$

C. $\frac{\lambda^2}{2}$

D. $\frac{\lambda^2}{4}$

Question 76:

Match the given columns:

	Column-I (Electrons in given subshell)		Column-II (Possible exchanges)
a.	$3d^4$	(i)	1
b.	$3p^2$	(ii)	3
c.	$2p^3$	(iii)	6
d.	$3d^5$	(iv)	10

Choose the correct match.

A. a(iii), b(i), c(iv), d(ii)

B. a(iii), b(i), c(ii), d(iv)

C. a(iv), b(iii), c(ii), d(i)

D. a(iv), b(i), c(iii), d(ii)

Question 77:

Choose the correct statement(s)
a. An orbital is designated by three quantum numbers while an electron is designated by four quantum numbers.
b. The electronic configuration of oxygen atom is



c. Stability of nitrogen atom can be explained by its half filled valence shell configuration.
Choose the correct option.

A. a and b only

B. c only

C. a and c only

D. a, b and c

Question 78:

Choose the sets with the species containing same number of atoms.
a. 10 g of O_3 and 10 g of O_2.
b. 1.12 mL of CO and 2.24 mL of CO_2 at STP.
c. N_A molecules of CS_2 and $3N_A$ atoms of He.
d. 5 moles of N_2 and 2 moles of NO_2.
Choose the correct option.

A. Sets a and c

B. Sets a, c and d

C. Sets b, c and d

D. Sets c and d

Question 79:

X and Y are the two elements which combine to form two different compounds XY_2 and X_3Y_2. If 0.5 mol of X_3Y_2 weighs 150 g and 0.2 mol of XY_2 weighs 40 g then the atomic weights of X and Y respectively are

A. 75 and 25

B. 25 and 75

C. 50 and 75

D. 75 and 50

Question 80:

Consider the following statements.
Statement I: Bronze is a homogeneous mixture while starch solution is heterogeneous.
Statement II: Homogeneous mixture has uniform composition throughout.
In the light of above statements, choose the correct option.

A. Both the statement I and statement II are correct

B. Both the statement I and statement II are incorrect

C. Statement I is correct but statement II is incorrect

D. Statement I is incorrect but statement II is correct

Question 81:

Consider the following pairs of species
a. CO_2 and CO
b. NO_2 and NO
c. CS_2 and CO_2
d. H_2O and H_2O_2
The pairs of compounds which can be considered as the examples of "Law of multiple proportions" are

A. a, b and c only

B. b, c and d only

C. a, b and d only

D. b and d only

Question 82:

If 5.85 g of sodium chloride is present in 250 mL of solution then the molarity of the solution will be

A. 0.6 M

B. 0.4 M

C. 0.8 M

D. 0.2 M

Question 83:

The volume of oxygen gas and the mass of 25% pure coal required respectively to produce 2.8 L of carbon dioxide gas on complete combustion at STP is

A. 1.4 L and 15 g

B. 2.8 L and 1.5 g

C. 1.4 L and 6 g

D. 2.8 L and 6 g

Question 84:

Choose the incorrect statement.

A. Mole fraction is a unitless concentration term

B. The sum of mole fractions of all the components in a solution is always unity

C. Both molality and mole fraction are temperature independent

D. Unit of molality is mol L^{-1}

Question 85:

Molarity of 24.5% (w/w) H_2SO_4 solution with density = 1.1 g mL^{-1}, is

A. 2.75 M

B. 2.30 M

C. 1.75 M

D. 2.05 M

Question 86:

What mass of 40% (w/w) pure $CaCO_3$ will be required to neutralise 25 mL of 0.5 M HCl solution according to following chemical reaction is
$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

A. 1.82 g

B. 1.56 g

C. 1.28 g

D. 1.12 g

Question 87:

Mole fraction of N_2 in an air sample if it contains 56% (w/w) nitrogen, 32% (w/w) oxygen and remaining argon (Atomic mass = 40 u), is

A. 0.3

B. 0.5

C. 0.6

D. 0.7

Question 88:

In a 1.5 molal aqueous solution of methanol, the mole fraction of methanol is

A. 0.084

B. 0.013

C. 0.039

D. 0.026

Question 89:

The molecule with empirical formula CH_3O is

A. Acetaldehyde

B. Ethylene glycol

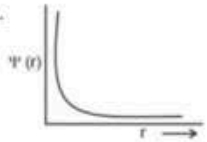
C. Acetic acid

D. Oxalic acid

Question 90:

Which of the given plots of wave function (Ψ) versus distance from the nucleus (r) represents 2*s* orbital?

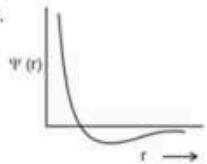
A.



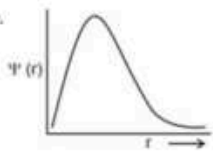
B.



C.



D.



Question 91:

Given organisms belong to how many families?

Wheat, Makoi, Potato, Housefly, Mango

A. Two

B. Three

C. Four

D. Five

Question 92:

Select the **correctly** written binomial epithet.

A. *Mangifera Indica*

B. *mangifera Indica*

C. Mangifera Indica

D. *Mangifera indica* Linn.

Question 93:

Read the given events of cell division and arrange them in a chronological order.
A. On equatorial plate, chromosomes appear four threaded
B. Occurrence of chromosome duplication
C. Chromatids of the univalent condenses
D. Two homologous chromosomes separate from each other
E. Terminalisation of chiasmata
Choose the **correct** option.

- A. E, A, B, C, D
- B. B, E, A, D, C
- C. B, C, E, A, D
- D. E, C, A, D, B

Question 94:

Which of the following belongs to the basic and lowest category in a taxonomic hierarchy?

A. Solanaceae

B. Sapindales

C. Dicotyledonae

D. *melongena*

Question 95:

Read the given statements.
A. The cellular structure first described by Robert Brown is the main site where glycogen is synthesized.
B. Some mature cells, like sieve tube cells and white blood cells, lack nucleus.
C. Telomeres and satellite play a role in preventing the chromosome loss because both are found at the ends of all chromosomes.
D. Gas vacuoles have unit membrane and provide buoyancy to the prokaryotic organisms found in aquatic habitat.
How many of the above given statements is/are **incorrect**?

- A. Two
- B. Three
- C. One
- D. Four

Question 96:

Systematics is

A. A term given by Flemming

B. Not associated with phylogeny

C. Derived from a Latin word

D. Derived from a Greek word

Question 97:

Read the below given features and identify the structure associated with it and choose the **correct** option.
a. Allows prokaryotic organisms to hide from host's immune system.
b. Associated with muscle contraction by release and uptake of calcium ions.
c. Help in formation of cleavage furrow during cell division in eukaryotes.
d. Occur in most of animals and plants but quite common in photosynthetic cells where these participates in photorespiration.

	a	b	c	d
(1)	Tonoplast	Sarcoplasmic reticulum	Phragmoplast	Microbodies
(2)	Cell wall	RER	Intermediate filaments	Plastids
(3)	Capsule	SER	Microfilaments	Peroxisomes
(4)	Glycocalyx	Dictyosomes	Microtubules	Mitochondria

A. (1)

B. (2)

C. (3)

D. (4)

Question 98:

Read the given statements and choose the **correct** option
Statement A: Duration of cell cycle can vary from organism to organism and also from cell type to cell type.
Statement B: Diplotene can lasts for months or years in oocytes of some vertebrates.

A. Only statement A is correct

B. Only statement B is correct

C. Both the statements A and B are true

D. Both the statements A and B are false

Question 99:

Round and biconcave shape is of

A. White blood cells

B. Mesophyll

C. Nerve cells

D. Human RBCs

Question 100:

Read the given statements and state them as **true(T)** or **false(F)** and choose the **correct** option.
A. Any sexually reproducing organism starts its life cycle from a single celled zygote.
B. Interkinesis is characterised by occurrence of centriole and chromosome duplication.
C. If a cell divides frequently then it has a longer G_1 phase and if a cell does not divide frequently, then it has shorter G_1 phase.

A. A-(F), B-(T), C-(T)

B. A-(T), B-(T), C-(T)

C. A-(T), B-(F), C-(F)

D. A-(F), B-(F), C-(T)

Question 101:

Select the **correctly** matched pair.

(1)	*Systema Naturae*	–	Written by Ernst Mayr
(2)	*Datura*	–	Placed in same family as that of wheat
(3)	Alsatian (dog)	–	Represent the organisms that belong to family Felidae
(4)	Human beings	–	Belong to the order primata

A. (1)

B. (2)

C. (3)

D. (4)

Question 102:

Mitotic apparatus is composed of

A. Spindle fibres and primary constriction

B. Centrosomes and kinetochores

C. Centromeres and filaments

D. Asters and spindle fibres

Question 103:

Match List-I with List-II and choose the **correct** option.

	List-I		List-II
a.	Chordata	(i)	Group of related orders
b.	Diptera	(ii)	Group of related classes
c.	Mammalia	(iii)	Group of related genera
d.	Hominidae	(iv)	It is a group of related families

A. a(iv), b(ii), c(iii), d(i)

B. a(ii), b(iv), c(i), d(iii)

C. a(i), b(iii), c(iv), d(ii)

D. a(i), b(iv), c(iii), d(ii)

Question 104:

Read the given **Assertion (A)** and **Reason (R)** statements and choose the **correct** option.
Assertion (A): Members of Convolvulaceae and Solanaceae are placed in same category Polymoniales.
Reason (R): Different plant families are included in the order Polymoniales mainly based on the vegetative characters.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true but (R) is not the correct explanation of (A)

C. (A) is true but (R) is false

D. Both (A) and (R) are false

Question 105:

The hypothesis '*Omnis cellula-e cellula*' was proposed by the scientist

A. Matthias Schleiden

B. Theodore Schwann

C. Nicolson

D. Rudolf Virchow

Question 106:

Read the given statements and choose the **correct** option.
Statement-A: *Familiaris*, *Aureus*, and *Pardus* are genera and all belong to same species.
Statement-B: Higher the category, greater is the difficulty of determining the relationship to other taxa at same level.

A. Both statements A and B are correct

B. Both statements A and B are incorrect

C. Only statement A is incorrect

D. Only statement B is incorrect

Question 107:

Select the **correctly** matched pair.

(1)	Quiescent stage	–	Cells are metabolically inactive
(2)	Asters	–	Made up of microtubules in prokaryotes
(3)	Male honey bees	–	Haploid cells divide by meiosis
(4)	Congression of chromosomes	–	Occurs at second stage of karyokinesis in equational division

A. (1)

B. (2)

C. (3)

D. (4)

Question 108:

Oxysomes are found to be associated with infoldings of membrane of an organelle. This organelle is

A. Lysosome

B. Ribosome

C. Mitochondrion

D. Peroxisome

Question 109:

The stage of karyokinesis in which chromosomes become thickest and shortest, is also characterised by

A. Splitting of centromere and separation of chromatids

B. Alignment of chromosomes along metaphasic plate through spindle fibres to both poles

C. Development of nuclear envelope around chromosome clusters at each pole

D. Beginning of centriole movement towards opposite poles

Question 110:

Select the **correct** statement.

A. The best stage to study shape of chromosomes in M-phase shows shortening of microtubules and separation of homologous chromosomes

B. If the DNA amount is 60 pg at G_2 phase, then each meiotic II product of this cell will contain 20 pg DNA

C. If APC (Anaphase promoting complex) become defective then separation of sister chromatids will be hampered

D. Diakinesis represents transition to metaphase II

Question 111:

A yeast cell differs from human cell as the former

A. Shows interphase that lasts more than 95% of the duration of cell cycle

B. Is devoid of 70S ribosome in semi-autonomous organelles

C. Completes its one cell cycle in about 90 minutes

D. Is an eukaryotic cell without cell wall

Question 112:

The important site of the formation of glycoproteins and glycolipids is

A. Smooth endoplasmic reticulum

B. Ribosome

C. Golgi apparatus

D. Rough endoplasmic reticulum

Question 113:

Select the **correctly** matched pair.

(1)	Pili	–	Provide motility to gram negative bacteria
(2)	Sap vacuoles	–	Absent in prokaryotic cell
(3)	Primary cell wall	–	Absent in parenchymatous cells
(4)	Residual bodies	–	Primary lysosomes containing digested substances

A. (1)

B. (2)

C. (3)

D. (4)

Question 114:

Read the given statements and choose the **correct** option.
Statement A: When the concentration of Mg^{2+} ions reduces below a critical level, both ribosomes subunits get separated from each other.
Statement B: Nucleoplasm is an opaque non-colloidal substance which fills the nucleus and contains nucleoprotein fibres.

A. Both statements A and B are correct

B. Both statements A and B are incorrect

C. Only statement A is incorrect

D. Only statement B is incorrect

Question 115:

The common feature between mesosome and mitochondria is that

A. Both are associated with 60S and 40S ribosomal subunits

B. Both can be found in all living organisms

C. Both help in the genomic DNA replication and distribution to daughter cells

D. Both are involved in the process of respiration

Question 116:

Acrocentric chromosome differs from submetacentric chromosome as the former

A. Possesses p and q arms

B. Appears to be L-shaped during third stage of karyokinesis of M-phase

C. Has one extremely short and one very long arm

D. Appears to have a single arm

Question 117:

Select the **odd** one out w.r.t. significance of mitosis.

A. Introduction of variations

B. Reproduction in unicellular organisms

C. Maintenance of cell size

D. Healing and regeneration

Question 118:

If all the daughter cells continue to divide, the number of mitotic generations required for producing 1024 cells from a cell would be

A. 6

B. 8

C. 10

D. 9

Question 119:

By the end of which of the following stages, the recombination between the homologous chromosome is completed?

A. Zygotene

B. Diplotene

C. Pachytene

D. Diakinesis

Question 120:

Phragmoplast is formed by

A. Plasma membrane

B. Fusion of lysosomes

C. Golgi complex

D. Microbodies

Question 121:

Select the **incorrect** statement.

A. Insects represent a group of organisms having three pairs of jointed legs.

B. Species is a group of potentially interbreeding populations.

C. Taxonomists have developed sub-categories in the taxonomic hierarchy to facilitate more sound and scientific placement of various taxa.

D. The rules of scientific naming of plants and animals is assigned in ICNCP.

Question 122:

The most dramatic period of cell cycle involve

A. Synthesis of tubulins and nucleotides

B. Duplication of centrosome and kinetochore

C. Duplication of genetic material in nucleus

D. A major reorganization of virtually all components of cell

Question 123:

Select the **odd** one out w.r.t. functions of plasma membrane.

A. Formation of intercellular junctions

B. Formation of glycoproteins

C. Endocytosis

D. Cell growth

Question 124:

Observe the given diagram and choose the **correct** option related to this stage of karyokinesis.



a. The structure, mainly involved in the rRNA synthesis disintegrates.
b. Shows reformation of reticular structures of endomembrane system.
c. In this stage nuclear membrane is assembled around the chromosomes.
d. Centrosome undergoes duplication during this stage and move towards opposite poles.
The **correct** one(s) is/are

A. b, c and d only

B. a and b only

C. All except b

D. a only

Question 125:

Which of the following is **not** found in prokaryotic cells?

A. Nucleoid

B. Peroxisome

C. Cyanophycean granules

D. Plasmids

Question 126:

Which of the following rule is **correct** w.r.t. binomial nomenclature?

A. The first word in a biological name starts with small letter.

B. Both the words in a biological name, when handwritten are underlined together

C. Biological names are generally in italics and written in Greek

D. Name of the author appears after specific epithet in abbreviated form

Question 127:

Identify the **correct** function of Golgi apparatus.

A. Replication of DNA

B. Synthesis of steroids and ATP

C. Osmoregulation and excretion

D. Formation of plasma membrane during cytokinesis

Question 128:

Match column-I with column-II.

	Column-I		Column-II
a.	Tetrad	(i)	Result of no cytokinesis after karyokinesis
b.	Syncytium	(ii)	Chiefly made up of calcium pectate
c.	Chiasmata	(iii)	Begins to appear in third stage of prophase I
d.	Middle lamella	(iv)	Becomes visible during fourth stage of prophase I

Choose the **correct** option.

A. a(ii), b(iv), c(iii), d(i)

B. a(iv), b(i), c(iii), d(ii)

C. a(iii), b(i), c(iv), d(ii)

D. a(iv), b(i), c(ii), d(iii)

Question 129:

Which of the following is **not** associated with the structure of centrosome?

A. Hub

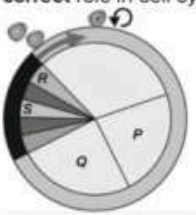
B. Radial spoke

C. Massule

D. Axoneme

Question 130:

Match the column-I with column-II w.r.t. the labelling given in the diagram and their **correct** role in cell cycle.



	Column-I		Column-II
a.	P	(i)	Synthesis of tubulin proteins and RNA
b.	S	(ii)	Distribution of mitochondria and plastids between daughter cells
c.	R	(iii)	Synthesis of basic histone proteins
d.	Q	(iv)	Formation of interzonal fibers

Choose the **correct** option.

A. a(iii), b(ii), c(i), d(iv)

B. a(ii), b(i), c(iv), d(iii)

C. a(iii), b(iv), c(ii), d(i)

D. a(i), b(iv), c(iii), d(ii)

Question 131:

Select the taxon associated with the genus *Musca*.

A. Chordata

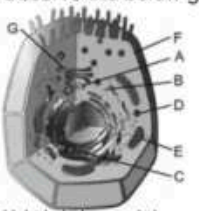
B. Primata

C. Mammalia

D. Arthropoda

Question 132:

Observe the below given diagram and choose the **correct** ones related to the labels.



(i) (C), (D) and (B) are single membrane bound organelles.
(ii) (A) is made up of nine evenly spaced peripheral doublet fibrils of tubulin protein.
(iii) (E) and (G) duplicate during G_2 phase of cell cycle.
(iv) (F) is the structure that is selectively permeable in nature and interacts with outside world.
(v) (C) provides precursors of enzymes for the formation of (D) in (G).

A. (i) and (iv) only

B. (iii) and (v) only

C. (ii), (iv) and (v) only

D. All except (ii)

Question 133:

According to IUCN, currently known and described species of all organisms are between

A. 2-50 million

B. 17-18 billion

C. 1.7-1.8 million

D. 9-20 billion

Question 134:

Read the given statements and state them as **true(T)** or **false(F)**.
A. Gibbon, tiger and *Canis* are placed in the order Carnivora along with *Felis*.
B. Mango, wheat and *Solanum* are placed in the same taxa, at class level.
C. Morphology, anatomy, along with cell structure, development processes, ecological and phylogenetic information of organisms forms the basis of modern taxonomic studies.
Choose the **correct** option.

A. A-(T), B-(F), C-(T)

B. A-(T), B-(T), C-(F)

C. A-(F), B-(F), C-(F)

D. A-(F), B-(F), C-(T)

Question 135:

Match column-I with column-II.

	Column-I		Column-II
a.	Primary constriction	(i)	Has mRNA and ribosomes
b.	Polysome	(ii)	Form spindle fibres
c.	Palade particles	(iii)	Splitting occurs during the third stage of karyokinesis in M-phase
d.	Centrioles	(iv)	Has rRNA and proteins

Choose the **correct** option.

A. a(iii), b(ii), c(i), d(iv)

B. a(ii), b(iv), c(iii), d(i)

C. a(iv), b(ii), c(i), d(iii)

D. a(iii), b(i), c(iv), d(ii)

Question 136:

All of the following are released *via* ducts onto a surface, **except**

A. Saliva

B. Sebum

C. Thyroxine

D. Pepsin

Question 137:

Complete the analogy by selecting the **correct** option.
Ciliated cuboidal epithelium : Bronchioles :: Simple squamous epithelium : _____.

A. Inner lining of pancreatic duct

B. Air sacs of lungs

C. Proximal convoluted tubule of nephron

D. Small intestine

Question 138:

_____ junction performs cementing between adjacent epithelial cells, preventing them from being pulled apart under mechanical stress.
Select the option that **correctly** fills the blank.

A. Communication

B. Gap

C. Synaptic

D. Adhering

Question 139:

'X' is the structure where maximum absorption of nutrients in alimentary canal occurs, while 'Y' is the structure where nearly all essential nutrients such as glucose as well as 70-80% of electrolytes from the tubular fluid in nephron are reabsorbed.
The common feature between the type of epithelium lining the 'X' and 'Y' is presence of

A. Microvilli on the free surface of cells

B. Tall and slender cells

C. Cells with basal nuclei

D. A single layer of flattened cells with irregular boundaries

Question 140:

Select the **correct** representation of palmitic acid.

A. $C_{14}H_{28}O_2$

B. $C_{16}H_{32}O_2$

C. $C_{18}H_{32}O_2$

D. $C_{20}H_{32}O_2$

Question 141:

Upon comparison between inulin and insulin, it was correctly concluded that

A. Both share the same type of peptide bonds

B. Latter one gives blue colour with iodine

C. Former one is a polymer of fructose

D. Latter one has a reducing end on its left side

Question 142:

Choose the **incorrect** statement w.r.t. enzymes.

A. Most enzymes are proteinaceous but ribozymes are RNA molecules with catalytic activity.

B. Co-factors are non-protein constituents that may be metal ions or organic groups.

C. A holoenzyme requires apoenzyme (a protein part) and co-factor for its activity.

D. Enzymes alter the potential energy of the substrate and equilibrium constant of the reactions they catalyse.

Question 143:

All of the following statements are **incorrect, except**

A. Glycogen is similar to starch as both of them are present as store house of energy in plant tissues.

B. Lipids are macromolecules with high molecular weight.

C. Cellulose is an unbranched polymer of glucose that is present in plant cell wall.

D. All nucleotides contain a deoxyribose sugar.

Question 144:

Respiratory tract of humans is composed of
(a) Epithelial tissue
(b) Cartilage
(c) Smooth muscle tissue
(d) Neural tissue
Choose the **correct** option.

A. (a) and (b) only

B. (a), (b), (c) and (d)

C. (b) and (c) only

D. (a), (b) and (c) only

Question 145:

Moist surface of buccal cavity is lined by 'X' epithelium which primarily helps/participates in

A. Forming a diffusion boundary

B. Secretion and absorption

C. Moving particles in a specific direction over the epithelium due to presence of cilia

D. Providing protection against chemical and mechanical stresses

Question 146:

Which of the following holds true for ligaments?

A. Attach skeletal muscle to bone

B. Form part of cartilage matrix

C. Example of dense connective tissue that connects one bone to another

D. Example of loose connective tissue that contains macrophages

Question 147:

Epithelial tissue performs
a. Secretion of digestive enzymes
b. Absorption of digested food
c. Sensory reception
Choose the **correct** option.

A. a, b and c

B. b and c only

C. a and b only

D. a and c only

Question 147:

Epithelial tissue performs
a. Secretion of digestive enzymes
b. Absorption of digested food
c. Sensory reception
Choose the **correct** option.

A. a, b and c

B. b and c only

C. a and b only

D. a and c only

Question 148:

Consider the given features:
a. Possesses compactly packed fibres and fibroblasts
b. Orientation of fibres shows an irregular pattern
c. Provides structural support
All of the above features are true for the type of tissue present in the

A. Endothelium

B. Skin

C. Myocardium

D. Endometrium

Question 149:

Select the **incorrect** statement:

A. Cardiac muscle fibres are striated, branched and uninucleated.

B. Smooth muscle fibres are spindle-shaped and lack striations.

C. Skeletal muscle fibres are striated, branched and multinucleated.

D. Smooth muscle fibres are involuntary in action.

Question 150:

Read the given statements and choose the **correct** option.
Statement A: Neuroglial cells make up less than half the volume of neural tissue in our body.
Statement B: Sponges exhibit division of labour at cellular level, despite of lacking true tissues.

A. Only statement A is correct

B. Only statement B is correct

C. Both statements A and B are correct

D. Both statements A and B are incorrect

Question 151:

Select the option that represents a **correct** difference between glucose and fructose.

	Glucose	Fructose
(1)	Pentose sugar	Hexose sugar
(2)	Contains a terminal aldehyde group	Contains a ketone group
(3)	Found only in plants	Found only in animals
(4)	Has 5 oxygen atoms	Has 7 oxygen atoms

A. (1)

B. (2)

C. (3)

D. (4)

Question 152:

Assertion (A): Enzymes are highly specific for their substrates.
Reason (R): The allosteric site of an enzyme has a unique 2-D shape complementary to the reaction's substrate.
In the light of the above statements, select the **correct** option.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true but (R) is not the correct explanation of (A)

C. (A) is true but (R) is false

D. (A) is false but (R) is true

Question 153:

A common feature between arachidonic acid and cytidylic acid is that both

A. Are found in acid-insoluble pool after chemical analysis of a living tissue

B. Contain phosphate group in their structure

C. Possess ester bonds in their structure

D. Have hydrogen in their structure

Question 154:

Match column I with column II.

	Column I		Column II
a.	Glycogen	(i)	Peptide bond
b.	Haemoglobin	(ii)	Glycosidic bond
c.	Triglyceride	(iii)	Phosphodiester bond
d.	DNA	(iv)	Ester bond

Choose the **correct** answer from the options given below.

A. a(i), b(ii), c(iii), d(iv)

B. a(iv), b(iii), c(ii), d(i)

C. a(iii), b(ii), c(i), d(iv)

D. a(ii), b(i), c(iv), d(iii)

Question 155:

Disaccharide $\xrightarrow{\text{'X'}}$ Two molecules of monosaccharides.
Choose the **correct** option w.r.t. the 'X' in above reaction.

A. 'X' represents the enzyme whose EC number begins with 4

B. The type of bond which is broken with the help of 'X', is also found in glucagon

C. 'X' is a hydrolase

D. 'X' is a dehydrogenase which belongs to the class I of enzymes

Question 156:

A medical student is examining a histological slide. She observes a single layer of cells that are closely packed with little intercellular matrix, resting on a thin non-cellular basement membrane. The tissue she is examining is involved mainly in

A. Providing structural support to the organs

B. Forming a lining over internal surfaces of hollow visceral organs

C. Exerting the greatest control over the body's responsiveness to changing conditions

D. Maintaining the positions of the various parts of the body

Question 157:

Assertion (A): Increasing the substrate concentration beyond a certain limit cannot increase the rate of an enzyme-catalysed reaction, even if the enzyme concentration as well as its optimum temperature and optimum pH are maintained.
Reason (R): At sufficiently high substrate concentration, all the active sites of the enzyme molecules become occupied, leaving no free enzyme molecules available to bind with additional substrate molecules.
In the light of above given statements, choose the **correct** option.

A. Both (A) and (R) are true and (R) is the correct explanation of (A).

B. Both (A) and (R) are true but (R) is not the correct explanation of (A).

C. (A) is true, but (R) is false.

D. (A) is false, but (R) is true.

Question 158:

Read the given statements and select the **correct** option.
Statement A: Osteocytes move freely between lamellae through large open channels called canaliculi and continuously remodel the hard and pliable matrix.
Statement B: The type of specialised connective tissue that has hard and non-pliable ground substance lacks any blood supply and is thus, non-vascular.

A. Both the statements A and B are correct

B. Both the statements A and B are incorrect

C. Only statement A is correct

D. Only statement B is correct

Question 159:

Choose the **incorrect** option w.r.t. areolar tissue.

A. Located beneath the skin and acts as a packing tissue

B. Contains collagen fibres, adipocytes and eosinophils

C. Its matrix, which accumulates between cells and fibres, is rigid and calcified

D. Provides a support framework for epithelium

Question 160:

Both α-helix and β-pleated sheet are representations of the ____ structures of proteins, stabilized primarily by ____ bonds formed between NH and C = O groups of the backbone.
Choose the option that **correctly** fills the respective blanks.

A. Primary : peptide

B. Tertiary : disulphide

C. Secondary : hydrogen

D. Quaternary : ionic

Question 161:

Select the **correct** match.

(1)	Lecithin	–	Possesses three fatty acid molecules linked to a phosphate group
(2)	Cholesterol	–	Do not contain any fatty acid
(3)	Alanine	–	Has an additional amino group
(4)	Thiamine	–	Heterocyclic structure present in DNA

A. (1)

B. (2)

C. (3)

D. (4)

Question 162:

The total number of oxygen atoms present in one molecule of glycerol is equal to the total number of

A. Nitrogen atoms in uracil

B. Hydrogen atoms in glycine

C. Heterocyclic rings in cytosine

D. Oxygen atoms in serine

Question 163:

Choose the **correct** statement.

A. All amino acids contain a sulfur atom in their R-group.

B. All proteins exhibit quaternary structure.

C. In proteins, only right handed helices are observed.

D. Enzymes work optimally only at neutral pH.

Question 164:

The most abundant protein in the animal world is

A. Present in dense irregular connective tissue and is a structural heteropolymer

B. Found in blood plasma and functions as a transport protein for CO_2

C. A globular receptor protein present on the surface of lymphocytes

D. A storage homopolymer found in the liver and skeletal muscle

Question 165:

Consider the given statements:
(a) Aspartic acid and palmitic acid are considered as biomacromolecules.
(b) On chemical analysis of a living tissue, the acid-insoluble pool represents roughly the cytoplasmic composition.
(c) Disulphide bonds are present in insulin as well as in an antibody molecule.
Choose the option with **incorrect** statements.

A. (a) and (b) only

B. (b) and (c) only

C. (a), (b) and (c)

D. (a) and (c) only

Question 166:

The type of epithelium present in the lining of our stomach is primarily involved in

A. Protection from alkaline pH

B. Secretion and absorption

C. Maintaining only sodium balance of body fluids

D. Forming a diffusion boundary for gaseous exchange

Question 167:

A biochemist compares the activity of four purified catalysts/biocatalysts under identical reaction conditions.

Catalyst/Biocatalyst	Nature	Source
I	Inorganic catalyst	Laboratory preparation
II	Biocatalyst	Human liver
III	Biocatalyst	Thermophilic bacterium isolated from a hot sulphur spring
IV	Biocatalyst	Mesophilic soil bacterium

Each catalyst is incubated at 85°C for one hour and then tested under its optimum pH with excess substrate.
Which of the following is most likely to be observed?

A. Biocatalysts II and IV will exhibit maximum catalytic activity because enzymes are more specific than inorganic catalysts.

B. Catalyst I and biocatalyst III are expected to retain catalytic activity, whereas II and IV are most likely to lose their catalytic activity.

C. Only catalyst I will remain catalytically active because all enzymes are irreversibly denatured above 40°C.

D. Catalyst I, Biocatalysts II, III, and IV will all show comparable catalytic activity because substrate concentration is in excess.

Question 168:

Identify the secretion among the following that is produced by a multicellular exocrine gland and select the **correct** option.

A. Mucus by goblet cells

B. Glucagon secretion by alpha cells of pancreas

C. Cortisol by adrenal gland

D. Saliva by salivary gland

Question 169:

Involuntary, branched and striated muscle fibres are characteristic of

A. Biceps

B. Walls of urinary bladder

C. Myocardium

D. Iris of the eye

Question 170:

Observe the following features:
(a) Solid and pliable matrix that resists compression
(b) Absence of blood vessels
(c) Matrix contains collagen and elastin fibres
This tissue is most likely

A. Bone

B. Cartilage

C. Tendon

D. Ligament

Question 171:

Select the **incorrect** match regarding cartilage.

(1)	Matrix component	–	Chondroitin sulphate in the ground substance
(2)	Cell type	–	Chondrocytes located in lacunae
(3)	Location	–	Outer ear joints, between adjacent bones in vertebral column
(4)	Structural feature	–	Concentric rings arranged around a central canal

A. (1)

B. (2)

C. (3)

D. (4)

Question 172:

The type of epithelium present in the inner most lining of blood vessels is characterised by all of the following features, **except**

A. Single layer of flat cells

B. Generally avascular

C. Nucleus of each cell is flattened and oval

D. Present at site where only filtration and protection is required

Question 173:

Magnesium ion acts as a co-factor in many phosphate-transfer enzymes. Which type of bond/interaction does this ion typically forms with the residues at the enzyme's active site?

A. Peptide bond

B. Ester bond

C. Co-ordination bond

D. Glycosidic bond

Question 174:

Choose the **correct** representation.

A. Nitrogenous base + Pentose sugar = Nucleotide

B. Nucleoside – Phosphate group = Nitrogenous base

C. Guanylic acid – Phosphate group = Guanosine

D. Cytidine + Phosphate group = Cytosine

Question 175:

How many of the biomolecules mentioned in the box below contain glycosidic bond(s) in their structure?

Chitin, Sucrose, Adenosine, Triglyceride, Uridylic acid, Cellulose, Starch

Select the **correct** option.

A. Six

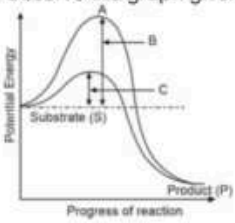
B. Four

C. Five

D. Three

Question 176:

Observe the graph given below and select the **correct** option w.r.t. it.



A. C represents activation energy in the absence of enzyme

B. The reaction depicted is an endothermic reaction since the potential energy of substrate is higher than the product

C. Point A represents the transition state, a transient, unstable and high energy intermediate in the reaction pathway

D. The enzyme lowers the activation energy by directly reducing the energy content of the product molecule

Question 177:

Malonate, a structural analog of succinate, inhibits the activity of succinic dehydrogenase. To overcome this inhibition experimentally, there should be

A. Increase in the concentration of malonate to saturate all inhibitor sites

B. Lowering of the temperature of the reaction to slow done malonate binding

C. Increase in the concentration of succinate

D. Addition of a non-competitive inhibitor to block the allosteric site and restore enzyme activity

Question 178:

Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Simple squamous epithelium	(i)	Covers the surface of ovary
b.	Compound epithelium	(ii)	Lines urinary bladder
c.	Simple cuboidal epithelium	(iii)	Lining of pharynx
d.	Transitional epithelium	(iv)	Bowman's capsule of kidneys

A. a(ii), b(i), c(iii), d(iv)

B. a(iv), b(iii), c(ii), d(i)

C. a(i), b(ii), c(iii), d(iv)

D. a(iv), b(iii), c(i), d(ii)

Question 179:

Choose the **odd** one w.r.t. structures possessing gap junctions.

A. Epithelial tissue

B. Electrical synapse

C. Cardiac muscles

D. Biceps

Question 180:

Select the **incorrect** option w.r.t. morphine.

A. Belongs to the category of alkaloids

B. Has ecological importance

C. Shares the same category of secondary metabolite as that of nicotine

D. Plays known role in normal physiological processes of all organisms